

EXPERIMENT 9

Aim: To implement private Blockchain using Geth.

Theory:

A **blockchain** is a distributed and decentralized ledger that records transactions securely across multiple nodes.

Geth (Go Ethereum) is the official command-line client used to run an Ethereum node. It allows you to:

- Create a private blockchain
- Mine blocks
- Deploy smart contracts
- Perform transactions

Private Blockchain

A **private blockchain** is a restricted network where:

- Only authorized participants can join
- It is controlled by a single organization or group
- Faster and more efficient than public blockchains

Key Concepts

- **Genesis Block**

The first block of the blockchain. It defines:

- Chain configuration
- Initial accounts
- Difficulty level

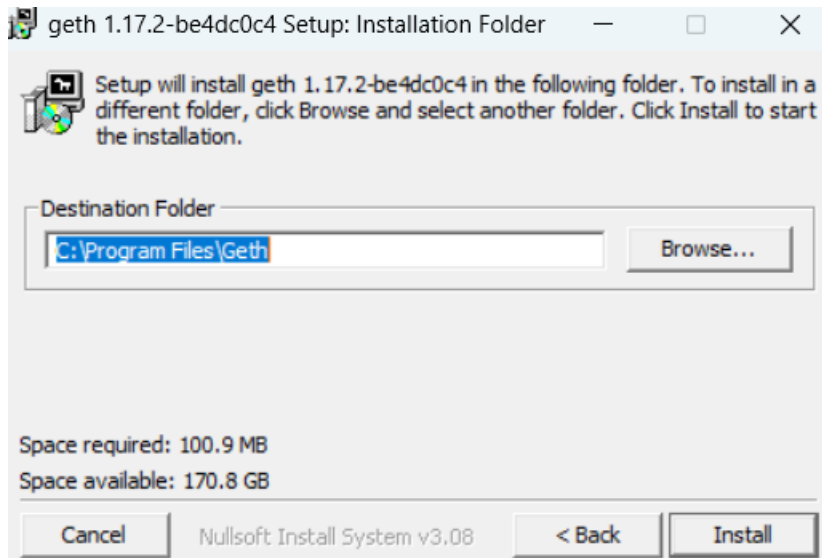
- **Node**
A system running Geth that participates in the blockchain network.
- **Mining**
The process of validating transactions and adding blocks to the chain.
- **Account**
Used to send/receive Ether in the network.

Steps to Implement Private Blockchain using Geth

1. Install Geth on your system (Windows/Linux).
2. Create a working directory for your private blockchain.
3. Create a genesis file to define the blockchain configuration (chain ID, difficulty, gas limit, etc.).
4. Initialize the blockchain using the genesis file.
5. Create a new account to use in the blockchain.
6. Start the Geth node with a custom network ID.
7. Open the Geth console to interact with the blockchain.
8. Start mining to generate blocks and Ether.
9. Check account balance to verify mining rewards.
10. Perform transactions between accounts.
11. View blockchain details like block number and block data.

Implementation:

Install Geth for windows from Ethereum website



Verify in command prompt if Geth is installed by using geth version

```
C:\Users\Archit>geth version
Geth
Version: 1.17.2-stable
Git Commit: be4dc0c4be2fe316dbdd0a73e48421f64978232f
Git Commit Date: 20260330
Architecture: amd64
Go Version: go1.25.7
Operating System: windows
GOPATH=
GOROOT=
```

Create project folder

```
C:\Users\Archit>mkdir private_blockchain

C:\Users\Archit>cd private_blockchain

C:\Users\Archit\private_blockchain>notepad genesis.json
```

Create a genesis.json file in notepad

C:\Users\Archit\private_blockchain>notepad genesis.json

```
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0
  },
  "difficulty": "2000",
  "gasLimit": "2100000",
  "alloc": {}
}
```

Initialize the private blockchain using geth

C:\Users\Archit\private_blockchain>notepad genesis.json

```
C:\Users\Archit\private_blockchain>geth --datadir data init genesis.json
INFO [04-10|07:01:21.361] Maximum peer count                ETH=50 total=50
INFO [04-10|07:01:21.369] Set global gas cap                 cap=50,000,000
INFO [04-10|07:01:21.370] Initializing the KZG library        backend=gokzg
INFO [04-10|07:01:21.372] Defaulting to pebble as the backing database
INFO [04-10|07:01:21.372] Allocated cache and file handles   database=C:\Users\Archit\private_blockchain\data\geth\
\chaindata cache=2.00GiB handles=8192
INFO [04-10|07:01:21.419] Opened ancient database            database=C:\Users\Archit\private_blockchain\data\geth\
\chaindata\ancient\chain readonly=false
INFO [04-10|07:01:21.419] Opened Era store                   datadir=C:\Users\Archit\private_blockchain\data\geth\
chaindata\ancient\chain\era
WARN [04-10|07:01:21.420] Head block is not reachable
INFO [04-10|07:01:21.420] State schema set to default        scheme=path
INFO [04-10|07:01:21.420] Load database journal from disk
INFO [04-10|07:01:21.433] Opened ancient database            database=C:\Users\Archit\private_blockchain\data\geth\
\chaindata\ancient\state readonly=false
INFO [04-10|07:01:21.435] State snapshot generator is not found
INFO [04-10|07:01:21.435] Starting snapshot generation       root=56e81f..63b421 accounts=0 slots=0 storage=0.00B
dangling=0 elapsed=0s
INFO [04-10|07:01:21.435] Initialized path database          triecache=16.00MiB statecache=16.00MiB buffer=64.00MiB
B state-history="last 90000 blocks" journal-dir=C:\Users\Archit\private_blockchain\data\geth\triedb
INFO [04-10|07:01:21.435] Resuming snapshot generation       root=56e81f..63b421 accounts=0 slots=0 storage=0.00B
dangling=0 elapsed=0s
INFO [04-10|07:01:21.435] Writing custom genesis block
INFO [04-10|07:01:21.435] Generated snapshot                 accounts=0 slots=0 storage=0.00B dangling=0 elapsed=0
```

Create an account

```
C:\Users\Archit\private_blockchain>geth --datadir data account new
INFO [04-10|07:01:37.578] Maximum peer count          ETH=50 total=50
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0xCC3dc783E0E473aa9fA9006EB95D84532c926Fa8
Path of the secret key file: data\keystore\UTC--2026-04-10T01-31-42.632300600Z--cc3dc783e0e473aa9fa9006eb95d845

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Start Geth Node

```
C:\Users\HOME\spandan\private_blockchain>geth --datadir data --networkid 1234 console
INFO [04-07|16:50:01.532] Maximum peer count          ETH=50 LES=0 total=50
INFO [04-07|16:50:01.595] Set global gas cap          cap=25000000
INFO [04-07|16:50:01.603] Allocated trie memory caches clean=256.00MiB dirty=256.00MiB
INFO [04-07|16:50:01.611] Allocated cache and file handles database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaindata
INFO [04-07|16:50:01.665] Opened ancient database      database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaindata\ancient
INFO [04-07|16:50:03.008] Initialised chain configuration config="{ChainID: 1234 Homestead: 0 DAO: <nil> DAOSupport: false EIP150: 0 EIP155: 0 EIP158: 0 Byzantium: 0 Constantinople: 0 Petersburg: 0 Istanbul: 0, Muir Glacier: <nil>, YOLO v2: <nil>, Engine: unknown}"
INFO [04-07|16:50:03.034] Disk storage enabled for ethash caches dir=C:\Users\HOME\spandan\private_blockchain\data\geth\ethash count=t=3
INFO [04-07|16:50:03.046] Disk storage enabled for ethash DAGs dir=C:\Users\HOME\AppData\Local\Ethash count=2
INFO [04-07|16:50:03.056] Initialising Ethereum protocol versions="[65 64 63]" network=1234 dbversion=<nil>
WARN [04-07|16:50:03.067] Upgrade blockchain database version from=<nil> to=8
INFO [04-07|16:50:03.077] Loaded most recent local header number=0 hash="6f14b1...a2805e" td=2000 age=57y1mo11h
INFO [04-07|16:50:03.091] Loaded most recent local full block number=0 hash="6f14b1...a2805e" td=2000 age=57y1mo11h
INFO [04-07|16:50:03.104] Loaded most recent local fast block number=0 hash="6f14b1...a2805e" td=2000 age=57y1mo11h
INFO [04-07|16:50:03.119] Regenerated local transaction journal transactions=0 accounts=0
INFO [04-07|16:50:03.141] Allocated fast sync bloom size=512.00MiB
INFO [04-07|16:50:03.148] Initialized fast sync bloom items=0 errorrate=0.000 elapsed=0s
INFO [04-07|16:50:03.149] Starting peer-to-peer node instance=Geth/v1.9.25-stable-e7872729/windows-amd64/go1.15.6
INFO [04-07|16:50:03.190] New local node record seq=1 id=eeee1fa666f7c09 ip=127.0.0.1 udp=30303 tcp=30303
INFO [04-07|16:50:03.206] IPC endpoint opened self=enode://a3e0ec411b9c6c658245a289eb62ae41f93a67c9839cf01d06840
INFO [04-07|16:50:03.319] Etherbase automatically configured url=\\.pipe\geth.ipc
address=0xa0c392e16b0f0e576683d2e5d51faFad2AcA6e4f
Welcome to the Geth JavaScript console!
```

Check accounts

```
> eth.accounts
["0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f"]
```

Unlock the account

```
> personal.unlockAccount(eth.accounts[0])
Unlock account 0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f
Passphrase: INFO [04-07|16:56:23.593] Looking for peers peercount=0 tried=66 static=0

true
```

Start Mining

```

> miner.start(1)
INFO [04-07|16:57:49.057] Updated mining threads          threads=1
INFO [04-07|16:57:49.063] Transaction pool price threshold updated price=1000000000
null
> INFO [04-07|16:57:49.070] Commit new mining work          number=1 sealhash="cb5512...b7dc49" uncles=0 txs=0 gas=0 fees=0 elapsed=1.007ms
INFO [04-07|16:57:50.937] Generating DAG in progress      epoch=0 percentage=0 elapsed=1.106s
INFO [04-07|16:57:52.040] Generating DAG in progress      epoch=0 percentage=1 elapsed=2.210s
INFO [04-07|16:57:53.133] Generating DAG in progress      epoch=0 percentage=2 elapsed=3.302s
INFO [04-07|16:57:54.226] Generating DAG in progress      epoch=0 percentage=3 elapsed=4.396s
INFO [04-07|16:57:55.338] Generating DAG in progress      epoch=0 percentage=4 elapsed=5.507s
INFO [04-07|16:57:55.981] Looking for peers               peercount=0 tried=59 static=0
INFO [04-07|16:57:56.461] Generating DAG in progress      epoch=0 percentage=5 elapsed=6.631s
INFO [04-07|16:57:57.583] Generating DAG in progress      epoch=0 percentage=6 elapsed=7.753s
INFO [04-07|16:57:58.706] Generating DAG in progress      epoch=0 percentage=7 elapsed=8.876s
INFO [04-07|16:57:59.803] Generating DAG in progress      epoch=0 percentage=8 elapsed=9.973s

```

INFO [04-07]20:45:12.391	Commit new mining work	number=2 sealhash="4f33f6...a9b2e0" uncles=0 txs=0 gas=0 fees=0 elap
sed="998μs"		
INFO [04-07]20:45:12.401	⚡ mined potential block	number=1 hash="e781cd...21e1bb"
INFO [04-07]20:45:12.916	Successfully sealed new block	number=2 sealhash="4f33f6...a9b2e0" hash="5590c8...45f96b" elapsed=524
.571ms		
INFO [04-07]20:45:12.916	Commit new mining work	number=3 sealhash="2fe669...f8004b" uncles=0 txs=0 gas=0 fees=0 elap
sed=0s		
INFO [04-07]20:45:12.926	⚡ mined potential block	number=2 hash="5590c8...45f96b"
INFO [04-07]20:45:15.455	Successfully sealed new block	number=3 sealhash="2fe669...f8004b" hash="c5becd...86cb9c" elapsed=2.5
39s		
INFO [04-07]20:45:15.455	Commit new mining work	number=4 sealhash="b6f543...2d8eb5" uncles=0 txs=0 gas=0 fees=0 elap
sed=0s		
INFO [04-07]20:45:15.466	⚡ mined potential block	number=3 hash="c5becd...86cb9c"
INFO [04-07]20:45:17.874	Successfully sealed new block	number=4 sealhash="b6f543...2d8eb5" hash="4bbde3...c3cd16" elapsed=2.4
19s		

Check account balance

```
> eth.getBalance(eth.accounts[0])  
2800000000000000000000  
> INFO [04-07|20:57:20.259] Looking for peers
```

Stop mining

```
> INFO [04-07|20:57:10.420] Looking for peers      peercount=0 tried=17 static=0
> miner.stop()
null
> INFO [04-07|20:57:52.482] Looking for peers      peercount=0 tried=43 static=0
INFO [04-07|20:58:02.489] Looking for peers      peercount=0 tried=50 static=0
INFO [04-07|20:58:12.708] Looking for peers      peercount=0 tried=18 static=0
```

To Send transaction

Created another account

```
C:\Users\HOME\spandan\private_blockchain>geth --datadir data account new
INFO [04-07|21:07:14.350] Maximum peer count          ETH=50 LES=0 total=50
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xc5Ab7B0f31c6304828eeC594b9ea37Af6D07f845
Path of the secret key file: data\keystore\UTC--2026-04-07T15-37-22.772205900Z--c5ab7b0f31c6304828eec594b9ea37af6d07f845

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Now we have 2 accounts

```
> eth.accounts
["0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f", "0xc5ab7b0f31c6304828eec594b9ea37af6d07f845"]
> INFO [04-07|21:09:10.220] Looking for peers peercount=0 tried=22 static=0
```

Unlock the account 0 and send transaction to account 1

```
> personal.unlockAccount(eth.accounts[0])
Unlock account 0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f
Passphrase: INFO [04-07|21:16:11.421] Looking for peers
peercount=0 tried=57 static=0

true
> INFO [04-07|21:16:22.171] Looking for peers
peercount=0 tried=40 static=0
> eth.sendTransaction({
..... from: eth.accounts[0],
..... to: eth.accounts[1],
..... value: web3.toWei(1, "ether")
..... })
INFO [04-07|21:16:31.327] Setting new local account
address=0xa0c392e16b0f0e576683d2e5d51FaFAd2AcA6e4f
INFO [04-07|21:16:31.337] Submitted transaction
fullhash=0x61629c85ab48b216143d2e288f5c6c30198fd6e81416315c516dcdb
5d7fa6982 recipient=0xc5Ab780f31c6304828eeC594b9eA37Af6D0f7845
"0x61629c85ab48b216143d2e288f5c6c30198fd6e81416315c516dcdb5d7fa6982"
> INFO [04-07|21:16:32.176] Looking for peers
peercount=0 tried=21 static=0
INFO [04-07|21:16:42.293] Looking for peers
peercount=0 tried=37 static=0
INFO [04-07|21:16:52.437] Looking for peers
peercount=0 tried=43 static=0
INFO [04-07|21:17:02.442] Looking for peers
peercount=0 tried=22 static=0
```

Start mining

```
miner.start(1)
INFO [04-07|21:19:03.349] Updated mining threads          threads=1
INFO [04-07|21:19:03.355] Transaction pool price threshold updated price=1000000000
INFO [04-07|21:19:03.361null] Com
mit new> mining work          number=23 sealhash="dcf750..5b31fd" uncles=0 txs=0 gas=0 fees=0 elapsed=0s
INFO [04-07|21:19:03.375] Commit new mining work          number=23 sealhash="960b1a..081554" uncles=0 txs=1 gas=21000 fees=2
.1e-05 elapsed=13.991ms
INFO [04-07|21:19:03.790] Looking for peers          peercount=0 tried=32 static=0
INFO [04-07|21:19:14.580] Looking for peers          peercount=0 tried=36 static=0
```

Stop mining

```
> miner.stop()
null
> INFO [04-07|21:21:36.229] Looking for peers
```

Account Balances

```
> eth.getBalance(eth.accounts[0])  
1130000000000000000000  
> INFO [04-07|21:24:18.456] Looking for peers peercount=0 tried=45 static=0  
> eth.getBalance(eth.accounts[1])  
1000000000000000000000
```

Transaction was successful as balance of account 0 has decreased and balance of account 1 is 1 ether which was transferred to account 1.

To fetch the details of the latest block mined

[illegible]

Checking the admin node info

```

admin.nodeInfo
{
  enode: "enode://a3e0ec411b9c6c658245a289eb62ae4a1f93a67c9839cf01d06840f25ea35bd73417385b2fe1b7c7fc18b1f11b097e6ac0981ace7aee219532c88296e496b93b049.43.24.168:30303",
  enr: "enr:-Je4QPpY4FL-2hvTHUIROVBWFA5vtBdPiMOX_zviaV0cVVt5hUajzdpP7ib6n-Uh_Ug49_GuSmJbhkBeILLiMj6DJD4Q0g2V0aMfGhI3Zu0uAgmlkgnY0gmLwhDExGkiJc2VjcdI1NmsxoQ0j40xBG5xsZYJFoonrYq5B-TpnyY0c8B0GhA8L6jW9c4N0Y3CCdL-DdWRwgnZf",
  id: "eeee1fa666f7c09061bfc3f965eab385282298806e745cd584c1b3ecdbf4261",
  ip: "49.43.24.168",
  listenAddr: "[::]:30303",
  name: "Geth/v1.9.25-stable-e7872729/windows-amd64/go1.15.6",
  ports: {
    discovery: 30303,
    listener: 30303
  },
  protocols: {
    eth: {
      config: {
        byzantiumBlock: 0,
        chainId: 1234,
        constantinopleBlock: 0,
        eip150Block: 0,
        eip150Hash: "0x0000000000000000000000000000000000000000000000000000000000000000",
        eip155Block: 0,
        eip158Block: 0,
        homesteadBlock: 0,
        istanbulBlock: 0,
        petersburgBlock: 0
      },
      difficulty: 7522001,
      genesis: "0x6f14b168d12c82428b52b485bdd66c77c8daf3c38dcc5c2678d03d260a2805e",
      head: "0xa2f12559d9f4bca9134f88ac1485816862f83f12e4e29d4a168bf0de50bf2cc6"
    }
  }
}

```

Conclusion:

A private blockchain was successfully implemented using Geth. The blockchain was initialized using a custom genesis block, and accounts were created for transactions. Mining was performed to generate Ether, and a transaction between accounts was successfully executed and verified. This demonstrates the working of a decentralized blockchain network in a controlled environment.